Student Course Management System

Source Code Submission

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Main.java

```
package SCM;

public class Main {

    public static void main(String[] args) {

        CourseManager manager = new CourseManager();

        manager.showMenu();

    }

}
```

Course.java

```
package SCM;

public class Course {

    private String courseCode;
    private String courseTitle;
    private int unit;

    public Course(String courseCode, String courseTitle, int unit) {
        this.courseCode = courseCode.trim().toUpperCase();
        this.courseTitle = courseTitle.trim();
        this.unit = unit;
    }

    public String getCourseCode() {
        return courseCode;
    }

    public String getCourseTitle() {
        return courseTitle;
    }

    public int getUnit() {
        return unit;
    }

    public void setCourseCode(String courseCode) {
        this.courseCode = courseCode.trim().toUpperCase();
    }

    public void setCourseTitle(String courseTitle) {
        this.courseTitle = courseTitle.trim();
    }

    public void setUnit(int unit) {
        this.unit = unit;
    }

    @Override
    public String toString() {
        return String.format("%-10s %-35s %d",
                courseCode,
                courseTitle,
                unit);
    }
}
```

CourseManager.java

```
package SCM;

import java.io.*;
import java.util.ArrayList;
import java.util.Scanner;

public class CourseManager {

    private ArrayList<Course> courseList = new ArrayList<>();
    private Scanner scanner = new Scanner(System.in);

    public void showMenu() {

        System.out.println("\n==============================");
        System.out.println(" Victor's Student Course Management System");
        System.out.println("==============================");
        System.out.println("1. Add Course");
        System.out.println("2. View Courses");
        System.out.println("3. Search Course");
        System.out.println("4. Compute Total Units");
        System.out.println("5. Save Courses");
        System.out.println("6. Load Courses");
        System.out.println("7. Exit");
        System.out.print("Enter your choice: ");

        try {

            int choice = Integer.parseInt(scanner.nextLine());

            switch (choice) {

                case 1:
                    addCourse();
                    break;

                case 2:
                    viewCourses();
                    break;

                case 3:
                    searchCourse();
                    break;

                case 4:
                    displayTotalUnits();
                    break;
```

```
                case 5:
                    saveCourses();
                    break;

                case 6:
                    loadCourses();
                    break;

                case 7:
                    System.out.println("Goodbye!");
                    return;

                default:
                    System.out.println("Invalid choice. Please enter a number between 1
and 7.");
            }

        } catch (Exception e) {
            System.out.println("Invalid input. Please enter a number.");
        }

        showMenu();
    }

    public void addCourse() {

        try {

            System.out.print("Enter Course Code: ");
            String code = scanner.nextLine().trim().toUpperCase();

            if(code.isEmpty()){
                throw new IllegalArgumentException("Course code cannot be empty.");
            }

            System.out.print("Enter Course Title: ");
            String title = scanner.nextLine().trim();

            if(title.isEmpty()){
                throw new IllegalArgumentException("Course title cannot be empty.");
            }

            System.out.print("Enter Course Unit: ");
            int unit = Integer.parseInt(scanner.nextLine());

            if(unit<=0){
                throw new IllegalArgumentException("Unit must be a positive number.");
            }

            courseList.add(new Course(code,title,unit));

            System.out.println("Course added successfully.");
```

```
        }catch(Exception e){

            System.out.println("Could not add course: "+e.getMessage());

        }

    }

    public void viewCourses(){

        if(courseList.isEmpty()){

            System.out.println("No courses have been added yet.");
            return;

        }

        System.out.println("\n-----------------------------------------------");
        System.out.printf("%-10s %-35s %s%n",
                "CODE","TITLE","UNIT");
        System.out.println("-----------------------------------------------");

        for(Course course : courseList){

            System.out.println(course);

        }

    }

    public void searchCourse(){

        System.out.print("Enter the Course Code to search for: ");

        String code = scanner.nextLine().trim().toUpperCase();

        Course foundCourse = findCourseByCode(code,0);

        if(foundCourse==null){

            System.out.println("No course found with code \""+code+"\".");

        }else{

            System.out.println("\nCourse found:");
            System.out.println(foundCourse);

        }

    }
```

```
    private Course findCourseByCode(String code, int index){

        if(index>=courseList.size())
            return null;

        if(courseList.get(index).getCourseCode().equalsIgnoreCase(code))
            return courseList.get(index);

        return findCourseByCode(code,index+1);

    }

    public void displayTotalUnits(){

        int totalUnits=0;

        for(Course course : courseList){

            totalUnits+=course.getUnit();

        }

        System.out.println("Total Units Across All Courses: "+totalUnits);

    }

    public void saveCourses(){

        try{

            BufferedWriter writer=new BufferedWriter(new FileWriter("courses.txt"));

            for(Course course : courseList){

                writer.write(course.getCourseCode()+","
                        +course.getCourseTitle()+","
                        +course.getUnit());

                writer.newLine();

            }

            writer.close();

            System.out.println("Courses saved successfully to courses.txt.");

        }catch(IOException e){

            System.out.println("Error saving courses to file.");

        }
```

```
    }
    public void loadCourses(){
        try{
            BufferedReader reader=new BufferedReader(new FileReader("courses.txt"));
            courseList.clear();
            String line;
            while((line=reader.readLine())!=null){
                String[] fields=line.split(",");
                if(fields.length==3){
                    courseList.add(
                            new Course(
                                    fields[0],
                                    fields[1],
                                    Integer.parseInt(fields[2])
                            )
                    );
                }
            }
            reader.close();
            System.out.println("Courses loaded successfully from courses.txt.");
        }catch(IOException e){
            System.out.println("No saved course file found.");
        }
    }
}
```